
Understanding high level behavioral contexts from egocentric videos

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The ability to understand context from a first-person perspective is crucial for a variety of emerging applications, including assistive technologies, robotics, and augmented reality. Egocentric videos, captured from wearable cameras, offer a unique viewpoint into daily human activities, reflecting real-world scenarios with inherent challenges such as motion blur, occlusions, and viewpoint shifts. This project proposes to develop a deep learning-based system for contextual scene understanding from egocentric video streams, moving beyond simple object or action recognition towards inferring higher-level contextual information about everyday life situations.

Our approach will leverage the rich visual and temporal information present in egocentric videos to identify and interpret the context of a scene. We hypothesize that by effectively modeling the relationships between objects, actions, and gaze patterns within a video sequence, we can achieve robust and nuanced context detection. Initially, we will focus on kitchen environments using the EPIC-Kitchens dataset [Damen et al., 2022], a large-scale resource with detailed annotations of objects, actions, and verbs. This will allow us to establish a strong baseline and experiment with different deep learning architectures, including Convolutional Neural Networks (CNNs) for visual feature extraction and Recurrent Neural Networks (RNNs) or Transformers for temporal modeling of video sequences. We will explore architectures capable of integrating multi-modal information, such as object presence and action sequences.

To expand beyond the kitchen domain and investigate the generalizability of our approach, we will incorporate the Charades-Ego dataset [Sigurdsson et al., 2018b]. Charades-Ego offers a broader range of indoor scenes and scripted activities, providing a complementary dataset to assess our model's ability to handle more diverse everyday contexts. We aim to develop a system capable of inferring not just the objects and actions present, but also the overarching context of the scene – for example, moving beyond recognizing "knife" and "vegetable" to understanding the context as "cooking preparation", or recognizing "computer" and "desk" within the context of "working in an office". This project will explore methods for hierarchical context representation, potentially moving from low-level visual cues to mid-level object-action relationships and finally to high-level contextual scene understanding. Ultimately, this research aims to contribute to a more comprehensive understanding of how deep learning can be used to interpret the rich contextual information embedded within egocentric video streams.

1 Literature Review

Recent research highlights the growing importance of egocentric video analysis for understanding human behavior. Li et al. [2021] address the challenge of training models for egocentric videos by proposing a transfer learning framework. Their "Ego-Exo" method uses third-person video datasets (like Kinetics-400 [Kay et al., 2017]) to pre-train models with auxiliary losses, improving performance on egocentric tasks in datasets such as Charades-Ego [Sigurdsson et al., 2018a] and EPIC-Kitchens-100 [Damen et al., 2022]. This work underscores the benefit of leveraging exocentric data to enhance egocentric video understanding. Ren and Philipose [2009] contribute a benchmark

dataset for egocentric object recognition, analyzing challenges specific to wearable cameras, such as motion blur and occlusions. Their analysis using a SIFT-based system quantifies these challenges and potential constraints, highlighting the unique difficulties of egocentric vision.

Moving beyond basic recognition, understanding context in egocentric videos is crucial. Jia et al. [2022] introduce the EgoTaskQA benchmark for evaluating task understanding in egocentric videos through question-answering. Their benchmark targets complex reasoning about action dependencies, intents, and agent beliefs, revealing gaps in current video reasoning models compared to human understanding of goal-oriented egocentric activities. This emphasizes the need for models that can perform higher-level contextual inference.

To address the complexity of long egocentric videos, Di and Xie [2024] propose a unified model for grounded question answering. Their approach integrates query grounding and answering, utilizing Large Language Models for data synthesis and introducing a closed-ended QA task for evaluation. Their GroundVQA model demonstrates state-of-the-art performance on benchmarks like QAEgo4D and Ego4D-NLQ, highlighting the effectiveness of unified training and temporal localization for question answering in extended egocentric video sequences.

2 Plan of Activities

1. Training backbone object/action detection models

We will start by training two baseline models which separately perform object detection and action classification. For each task, we will experiment with simpler models trained from scratch and evaluate their performances in comparison to those of state-of-the-art models that are fine-tuned using the EPIC-Kitchens-100 and Charades-Ego datasets.

- **Object Detection (Umberto):** Development and evaluation of object detection models, including simpler architectures such as RCNNs and state-of-the-art YOLO models [Redmon and Farhadi, 2018].
- **Action Classification (Jin-seo):** Implementation of 3D CNNs and other baseline architectures for action recognition.

The models with the best performance (that most closely resembles the comparison state-of-the-art) while minimizing complexity will be chosen as our final backbone architecture.

2. Inferring higher-level behaviors (Jin-seo)

Based on the object and action information extracted from these initial backbones, we will place additional layers on top that can predict high level behavioral context using these low-level features. Architectures that can encode temporal relations, such as RNNs, LSTM, and Transformers will be considered and tested. The layer will take in a sequence of detected objects as well as action-related visual features extracted from backbone models as input, and leverage the temporal relationship as additional cues to infer final high level behavior.

3. Final performance evaluation & Qualitative analysis (Umberto)

We will evaluate our model’s generalizability and overall performance on the held-out test set extracted from a subset of both EPIC-Kitchens-100 and Charades-Ego datasets. Qualitative analysis will also be performed to assess in which circumstances our model makes accurate and inaccurate higher level behavior predictions.

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